IC Selection: LM5164 Vs. LMR33630

Neither converter is “universally” better than another; it mostly depends on the situational demands. Thus, I’m defining the use case for this buck converter as a converter powering a PER sensor and data-acquisition PCB containing a microcontroller, CAN transceiver, and analog sensors near the drivetrain. Since this board operates near electrically noisy, high-current systems and may experience transient loads as sensors are sampled and CAN messages are transmitted, I prioritized efficiency, transient response, output ripple, EMI, and PCB space consumption. Both converters satisfy basic electrical requirements for this challenge such as I/O voltage requirements and suitable output current, so my selection focuses on optimally suiting this specific application.
Datasheet Comparison (LM5164 Vs. LMR33630)

Criterion	Goal	LM5164	LMR33630	Justification
Efficiency @ Operating Point	Maximize efficiency near 12-17V → 5V, 0.5A to reduce wasted power and heat on powered PCB	High efficiency; for specific operating point, no direct values	Datasheet includes 5 V efficiency curves and states >95% peak efficiency	Conditions don’t match fully, so can’t conclusively say which is better, but important IC context
MOSFET Resistance	Minimize internal switch resistance to reduce power dissipated as $P = I^2R$	HS: 725 mΩ; LS: 330 mΩ	VQFN HS: 75 mΩ; LS: 50 mΩ	LMR33630 wins. Its lower MOSFET resistance reduces conduction loss, as load current rises.
EMI / Switching Noise	Switching MOSFET → rapid current/voltage changes → high frequency electrical noise → goal is minimizing this noise	Explicitly optimized for ultra-low EMI; CISPR 25 Class 5 stated (lowest EMI classification)	Datasheet states low EMI and low switching noise; HotRod package too	LM5164 advantage in documented EMI validation.
Switching Frequency	higher frequency can shrink L and C, but generally increases switching loss	Adjustable up to 1 MHz	Fixed variants at 400 kHz, 1.4 MHz, or 2.1 MHz	LMR33630 advantage for compact implementations; LM5164 advantage in adjustability.
Package / PCB Area	Minimize footprint to preserve board space while maintaining good routing and thermal performance	SO PowerPAD: 4.9 × 6 mm	VQFN: 3 × 2 mm	LMR33630 advantage. Its VQFN package has a much smaller nominal footprint
Cost	Minimize cost on BOM (lower will win)	~$4.50 for singular	~$3 cost	LMR 33630 seems better, but might be able to find LM5164 at Detkin

Input Voltage, Output Voltage, and Output current are relatively similarly qualified for both converters, so I wanted to more heavily consider other equally important parameters.

Transient response and output ripple matter too because sudden MCU/CAN load changes can cause voltage droop or overshoot, while switching ripple can introduce noise into analog sensor measurements. Neither provides a clear winner because both depend heavily on external component selection, although the LMR33630’s higher switching-frequency options can meet ripple targets with smaller passive components, supporting a more compact PCB.

Final Selection: Overall, I recommend the **LMR33630** for this application. Its lower MOSFET resistance reduces conduction losses, while its smaller VQFN package and higher switching frequencies allow for a more compact PCB. The LM5164 does have an important advantage with its documented CISPR 25 Class 5 EMI performance, especially given that this board would operate near sensitive sensors and drivetrain electronics. However, since both converters are designed with low EMI in mind, I think the LMR33630's lower conduction resistance, smaller footprint, and lower cost outweigh this advantage for my specific use case. The final design would still need appropriate inductor and capacitor values to meet the ±20 mA current-ripple and ±50 mV voltage-ripple requirements.